FURTHER MECHANICS / KINEMATICS & DYNAMICS

Impulse and Work Done by a Variable Force

Mark scheme 11 questions · 123 marks

EXAM BOARDS AQA 7367/3M · OCR A Y543

Questions

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	Total	**123**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes. A grey line gives the rule where the name alone doesn't.

ft	follow through Correct work from the student's own earlier wrong value earns the mark.	**AG**	answer given The answer is printed in the question, so every step to it must be shown.
cao	correct answer only No follow-through from an earlier error.	**cso**	correct solution only No error anywhere in the part, not just in its answer.
oe	or equivalent	**awrt**	answers which round to
soi	seen or implied Earned if written, or if later working shows it was used.	**isw**	ignore subsequent working A correct answer, once reached, isn't spoiled by later working.
SC	special case Credit, as the note says, for work outside the main method.	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- **[*n*]** ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of g

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1(a)	$\mathbf{I} = 0.40(12\mathbf{j}) - 0.40(9\mathbf{i})$	**M1**	Uses $m(\mathbf{v} - \mathbf{u})$ with both velocities as vectors; condone the subtraction reversed. Not for subtracting speeds.
	$= (-3.6\mathbf{i} + 4.8\mathbf{j})\,\mathrm{N\,s}$	**A1**	oe, such as a column vector.
	$\lvert\mathbf{I}\rvert = \sqrt{(-3.6)^2 + 4.8^2}$	**M1**	Uses Pythagoras on both components of their impulse vector.
	$= 6\,\mathrm{N\,s}$	**A1**	
		[4]	
1(b)	$\int_0^{0.040} k\sin(25\pi t)\,\mathrm{d}t =$ "6" $\Rightarrow \left[-\frac{k}{25\pi}\cos(25\pi t)\right]_0^{0.040} =$ "6"	**M1**	Equates the integral of F over $0 \leq t \leq 0.040$ to their magnitude from (a) and integrates to the form $p\cos(25\pi t)$. Condone sign and coefficient slips.
	$\frac{k}{25\pi}(\cos 0 - \cos \pi) = \frac{2k}{25\pi}$	**A1**	Correct integral with both limits applied, oe.
	$k = 75\pi$	**A1**	cao
		[3]	
Total for Question 1		**7**	

Question	Scheme	Marks	Notes
2(a)	$I = 6(0) - 6(4)\ [= -24]$	**M1**	Uses impulse as change in momentum with mass 6 and speeds 4 and 0. Condone a sign slip.
	$24\,\text{N s}$	**A1**	Must be positive.
		[2]	
2(b)	$\int_0^T 18\,000t(T-t)\,\mathrm{d}t =$ "24" $\Rightarrow 18\,000\left[\frac{Tt^2}{2} - \frac{t^3}{3}\right]_0^T =$ "24"	**M1**	Equates the integral of F from 0 to T to their impulse from (a) and integrates, at least one power increased by 1. Condone sign slips.
	$3000T^3 = 24$	**A1**	Correct integral, $3000T^3$ oe.
	$T = 0.2$	**A1**	
	$\frac{\mathrm{d}F}{\mathrm{d}t} = 18\,000(T-2t) = 0 \Rightarrow t = \ldots$	**M1**	Differentiates F, at least one power reduced by 1, and sets it to zero, or uses the quadratic's symmetry, to find t in terms of T or their T.
	$F_{\max} = 18\,000(0.1)(0.2-0.1) = 180\,\text{N}$	**A1**	
		[5]	
	Total for Question 2	**7**	

Question	Scheme	Marks	Notes
3(a)	$W = \int_0^{0.5} \left(\frac{A}{1+2x} + 200x\right) \mathrm{d}x = \ldots$	**M1**	Integrates F with respect to x between 0 and 0.5, at least one term integrated correctly. Limits may be applied later.
	$\left[\frac{A}{2}\ln(1+2x) + 100x^2\right]_0^{0.5}$	**A1**	Both terms correct; logarithm term oe, such as $\frac{A}{2}\ln\left(x+\frac{1}{2}\right)$.
	$W = \frac{A}{2}\ln 2 + 25$ (J)	**A1**	oe, such as $A\ln\sqrt{2} + 25$, or awrt $0.347A + 25$.
		[3]	
3(b)	$\frac{1}{2}(20)(3)^2 + (20)(9.8)(0.5\sin 30°)$ $-$ "$\frac{A}{2}\ln 2 + 25$" $= 0$	**M1 A1**	**M1** Forms an energy equation with initial KE, GPE using a resolved drop, and their (a) of the form $pA + q$. Condone sign slips and sin/cos confusion. **A1** Correct equation, with (a) correct.
	$A = \frac{228}{\ln 2}$	**A1**	Any exact form, such as $\frac{456}{\ln 4}$.
		[3]	
	Total for Question 3	**6**	

Question	Scheme	Marks	Notes
4(a)	$I - 40 \times 2 = 80(7 - 3)$	**M1 A1**	**M1** Forms an impulse–momentum equation with the launcher's impulse, the resistance's impulse 40×2 and the change in momentum. Condone sign slips. **A1** Correct equation, oe.
	$I = 400\,\text{N s}$	**A1**	
		[3]	
4(b)(i)	"400" $= \int_0^2 kt^2(2 - t)\,\mathrm{d}t$ $= k\left[\frac{2t^3}{3} - \frac{t^4}{4}\right]_0^2$	**M1**	Equates $\int_0^2 F\,\mathrm{d}t$ to their impulse from (a) and integrates, the power of at least one term increased by 1.
	$= \frac{4k}{3}$	**A1**	Correct integral with limits applied, oe.
	$k = 300$	**A1**	
		[3]	
4(b)(ii)	$\frac{\mathrm{d}F}{\mathrm{d}t} =$ "300"$(4t - 3t^2) = 0 \Rightarrow t = \ldots$ $[= \frac{4}{3}]$ $F =$ "300" $\left(\frac{4}{3}\right)^2\left(2 - \frac{4}{3}\right)$	**M1**	Differentiates their F, the power of at least one term decreased by 1, sets it to zero and substitutes a nonzero root into F.
	$F_{\max} = \frac{3200}{9}\,\text{N}$	**A1**	$\frac{3200}{9}$ or awrt 356.
		[2]	
	Total for Question 4	**8**	

Question	Scheme	Marks	Notes
5(a)	$I = \int_0^2 (3t^2 - 12t + 9)\,\mathrm{d}t$	**M1**	Integrates F with respect to t between 0 and 2, with at least one power increased by 1.
	$= \left[t^3 - 6t^2 + 9t\right]_0^2$	**A1**	
	$I = 2\,\mathrm{N\,s}$	**A1**	
		[3]	
5(b)	"2" $= 2v - 2(3) \Rightarrow v = \ldots$	**M1**	Uses impulse = change in momentum with their impulse from (a), mass 2 and initial velocity 3.
	$4\,\mathrm{m\,s^{-1}}$	**A1ft**	ft $3 + \frac{1}{2}$ their impulse from (a).
		[2]	
5(c)	$3t^2 - 12t + 9 = 0 \Rightarrow t = 1, 3$ Force opposes the motion for $1 < t \leq 2$	**B1**	Interval from $t = 1$ to $t = 2$ identified; strictness of inequalities condoned.
	$\int_0^1 (3t^2 - 12t + 9)\,\mathrm{d}t = 2v_1 - 2(3) \Rightarrow v_1 = \ldots$	**M1**	Equates an integral of F over 0 to 1 (or 1 to 2, with their speed at $t = 2$) to the change in momentum. Condone slips.
	$v_1 = 5\,\mathrm{m\,s^{-1}}$	**A1**	
	$\frac{1}{2}(2)\,(\text{"5"})^2 + W = \frac{1}{2}(2)\,(\text{"4"})^2$	**M1**	Equates the work done to the change in KE between $t = 1$ and $t = 2$, with their speeds at both times. Condone sign slips.
	$W = -9\,\mathrm{J}$	**A1**	$-9\,\mathrm{J}$, or $9\,\mathrm{J}$ stated as work done against the motion.
		[5]	
	Total for Question 5	**10**	

Question	Scheme	Marks	Notes
6(a)	$I = \int_0^2 (3t^2 - 12t)\,\mathrm{d}t$ $= \left[t^3 - 6t^2\right]_0^2$	**M1 A1**	**M1** Integrates F with respect to t, at least one power increased by 1, and uses the limits 0 and 2. **A1** Correct integral, oe.
	$I = -16 \Rightarrow$ magnitude $16\,\mathrm{N\,s}$	**A1**	A0 if the magnitude is left negative.
		[3]	
6(b)	$4\frac{\mathrm{d}v}{\mathrm{d}t} = 3t^2 - 12t \Rightarrow 4v = t^3 - 6t^2 + c$	**M1**	Uses $F = ma$ with $a = \frac{\mathrm{d}v}{\mathrm{d}t}$, or impulse equals change in momentum, and integrates F with respect to t. Condone a missing constant.
	$v = 4,\ t = 0 \Rightarrow 4v = t^3 - 6t^2 + 16$	**A1**	oe, such as $v = \frac{1}{4}t^3 - \frac{3}{2}t^2 + 4$, or $-27 = 4v - 16$ from the impulse over $0 \leq t \leq 3$.
	$t = 3 \Rightarrow v = -\frac{11}{4}$ Speed $\frac{11}{4}\,\mathrm{m\,s^{-1}}$, negative direction	**A1**	Speed $\frac{11}{4}$ oe, given as positive, and the negative direction stated.
		[3]	
6(c)	"$t^3 - 6t^2 + 16$" $= 0$	**M1**	Sets their v from (b), a cubic in t, equal to zero.
	$(t-2)(t^2 - 4t - 8) = 0$	**dM1 A1**	**dM1** Takes out the factor $(t-2)$ by division, inspection or comparing coefficients, reaching a quadratic factor. **A1** Correct quadratic factor.
	$t^2 - 4t - 8 = 0 \Rightarrow t = 2 \pm 2\sqrt{3}$ $t = 2 + 2\sqrt{3}\,\mathrm{s}$	**A1**	Exact, oe such as $2 + \sqrt{12}$. A0 if $2 - 2\sqrt{3}$ is also given as an answer.
		[4]	
	Total for Question 6	**10**	

Question	Scheme	Marks	Notes
7(a)	$I = \int_0^1 (6t^2 - 24t + 18)\,\mathrm{d}t$ $= \left[2t^3 - 12t^2 + 18t\right]_0^1$	**M1 A1**	**M1** Integrates F with respect to t, the power of at least one term increased by 1, with limits 0 and 1 used. **A1** Correct integral, oe.
	$8\,\mathrm{N\,s}$	**A1**	
		[3]	
7(b)	"8" $= 4v - 4(-1) \Rightarrow v = \ldots$	**M1**	Uses impulse equals change in momentum with $m = 4$, $u = -1$ and their impulse from (a).
	$v = 1\,\mathrm{m\,s^{-1}}$	**A1**	
		[2]	
7(c)	$4\frac{\mathrm{d}v}{\mathrm{d}t} = 6(t-1)(t-3) \Rightarrow v = \ldots$	**M1**	Divides by the mass and integrates with respect to t, the power of at least one term increased by 1.
	$v = \frac{1}{2}t^3 - 3t^2 + \frac{9}{2}t + c$	**A1**	
	"1" $= \frac{1}{2} - 3 + \frac{9}{2} + c \Rightarrow c = \ldots$	**dM1**	Uses their velocity at $t = 1$ from (b) to find c; a definite integral from $t = 1$ is equivalent.
	$v = \frac{1}{2}t^3 - 3t^2 + \frac{9}{2}t - 1$	**A1**	oe
	$v = 1$: $t^3 - 6t^2 + 9t - 4 = 0 \Rightarrow t = \ldots$ $v = -1$: $t^3 - 6t^2 + 9t = 0 \Rightarrow t = \ldots$	**M1**	Sets their cubic v equal to 1 and to -1 and solves both, by any means.
	$t = 1,\ 3,\ 4$	**A1**	These three only; A0 if $t = 0$ is given as an answer.
		[6]	
	Total for Question 7	**11**	

Question	Scheme	Marks	Notes
8(a)	$W = \int_0^h \left(498 + \frac{k}{(x+2)^2}\right) \mathrm{d}x$ $= \left[498x - \frac{k}{x+2}\right]_0^h$	**M1 A1**	**M1** Integrates P with respect to x, at least one term correct, and uses limits 0 and h. **A1** Correct integral, oe; limits need not be substituted yet.
	$= 498h + \frac{k}{2} - \frac{k}{h+2}$	**A1**	oe, such as $498h + \frac{kh}{2(h+2)}$.
	$498 + \frac{k}{(x+2)^2} = 60(9.8) \Rightarrow x = \ldots$	**M1**	Sets P equal to the weight 588 and rearranges for x. Condone slips in the rearrangement.
	$x = \sqrt{\frac{k}{90}} - 2$	**A1**	oe. Positive root only.
		[5]	
8(b)	"$498x + k\left(\frac{1}{2} - \frac{1}{x+2}\right)$" $= \frac{1}{2}(60)v^2 + 588x$	**M1**	Forms an energy equation, at general x or at $x = 6$, with their work from (a) in terms of k and x, KE and the GPE gained. Condone sign slips.
	$30(3)^2 = -90(6) + k\left(\frac{1}{2} - \frac{1}{8}\right)$	**A1**	oe.
	$k = 2160$	**A1**	
	$30v^2 = -90(8) +$ "2160" $\left(\frac{1}{2} - \frac{1}{10}\right)$ $\Rightarrow v = \ldots$	**dM1**	Substitutes $x = 8$ and their k into their energy equation and solves for v.
	$v = 2.19\,\mathrm{m\,s^{-1}}$	**A1**	awrt 2.19.
		[5]	
8(c)	$0 = -90h +$ "2160" $\left(\frac{1}{2} - \frac{1}{h+2}\right)$ $\Rightarrow h = \ldots$	**M1**	Sets $v = 0$ in their energy equation from (b), with their k, and solves for a nonzero h.
	$h = 10\,\mathrm{m}$	**A1**	
		[2]	
8(d)	Small frontal area relative to its mass, so air resistance is small compared with its weight	**E1**	Gives a reason why air resistance is small beside the weight, such as small frontal area relative to mass, high density or low speed. Not for "compact" alone.
		[1]	
	Total for Question 8	**13**	

Question	Scheme	Marks	Notes
9(a)	$I = k\int_0^{0.1}(0.2t - t^2)\,\mathrm{d}t$ $\quad + k\int_{0.1}^{0.2}(0.2 - t)^2\,\mathrm{d}t$ $= k\left[0.1t^2 - \frac{t^3}{3}\right]_0^{0.1}$ $\quad + k\left[0.04t - 0.2t^2 + \frac{t^3}{3}\right]_{0.1}^{0.2}$	**M1 A1**	**M1** Splits the integral at $t = 0.1$ and integrates both pieces, at least one term of each increased in power. Condone a missing k. **A1** Both correct, oe such as $-\frac{(0.2-t)^3}{3}$.
	$= k\left(0.001 - \frac{0.001}{3}\right) + k\left(\frac{0.008}{3} - \frac{0.007}{3}\right)$	**dM1**	Substitutes both limits into each of their integrated pieces and subtracts.
	$= \frac{k}{1500} + \frac{k}{3000}$	**A1**	
	$= \frac{k}{1000}\,\mathrm{N\,s}$	**A1 AG**	Correct integrals and limits leading without error to $\frac{k}{1000}$. Numerical integration alone earns nothing.
		[5]	
9(b)	$u = 5\sin 60° \quad \left[= \frac{5\sqrt{3}}{2}\right]$	**B1**	Perpendicular approach component, oe such as $5\cos 30°$, either sign.
	$\frac{k}{1000} = 0.4\left(e \times \text{“}\frac{5\sqrt{3}}{2}\text{”}\right)$ $\quad + 0.4\left(\text{“}\frac{5\sqrt{3}}{2}\text{”}\right)$	**M1**	Equates the impulse from (a) to the change in perpendicular momentum, with rebound component e times their approach component. Condone sign slips.
	$k = 1000\sqrt{3}(e + 1)$	**A1**	oe.
	k increases with e, so $1000\sqrt{3}\left(1 + \frac{1}{4}\right) \le k \le 1000\sqrt{3}\left(1 + \frac{3}{4}\right)$	**dM1**	Substitutes $e = \frac{1}{4}$ and $e = \frac{3}{4}$ into their linear k in e, or carries the bounds on e through to k, or makes e the subject and applies its bounds.
	$1250\sqrt{3} \le k \le 1750\sqrt{3}$	**A1**	Exact bounds oe, both inequalities non-strict, justified by k increasing with e (a positive coefficient) or by the bounds carried through.
		[5]	
9(c)	$1500\sqrt{3} = \text{“}1000\sqrt{3}(e + 1)\text{”}$ $\Rightarrow e = \ldots \quad \left[= \frac{1}{2}\right]$ $v = e \times \text{“}\frac{5\sqrt{3}}{2}\text{”}$	**M1**	Finds the perpendicular rebound component, from e by their k in terms of e from (b), or directly from impulse $1.5\sqrt{3}$ equal to the change in momentum.
	$v = \frac{5\sqrt{3}}{4}$	**A1**	oe, or awrt 2.17.

Question 9 continued

Question	Scheme	Marks	Notes
	$V = \sqrt{\left(\text{“}\frac{5\sqrt{3}}{4}\text{”}\right)^2 + (5\cos 60°)^2}$ $\tan\theta = \text{“}\frac{5\sqrt{3}}{4}\text{”} \div 5\cos 60°$	**M1**	Uses Pythagoras for the speed, or a trigonometric ratio for the angle, with their perpendicular component and the parallel component unchanged by the impact. Condone sin/cos confusion.
	$V = \frac{5\sqrt{7}}{4}\,\text{m}\,\text{s}^{-1}$	**A1**	Exact oe, such as $\frac{\sqrt{175}}{4}$.
	$\theta = 40.9°$	**A1**	awrt 40.9°, measured from the wall.
		[5]	
	Total for Question 9	**15**	

Question	Scheme	Marks	Notes
10(a)	$800v\frac{\mathrm{d}v}{\mathrm{d}x} = 1800 + 1200e^{-x/200} - 2v^2$	**M1 A1**	**M1** Uses Newton's second law with $a = v\frac{\mathrm{d}v}{\mathrm{d}x}$, the driving force and the resistance. Condone sign slips. **A1** Correct equation, oe.
	$\frac{\mathrm{d}u}{\mathrm{d}x} = 2v\frac{\mathrm{d}v}{\mathrm{d}x}$ $\Rightarrow 400\frac{\mathrm{d}u}{\mathrm{d}x} = 1800 + 1200e^{-x/200} - 2u$	**M1 A1**	**M1** Replaces $v\frac{\mathrm{d}v}{\mathrm{d}x}$ by a multiple of $\frac{\mathrm{d}u}{\mathrm{d}x}$ and v^2 by u, reaching an equation in u and x only. Condone a wrong multiple. **A1** Correct equation in u and x, oe.
	$\frac{\mathrm{d}u}{\mathrm{d}x} + \frac{u}{200} = \frac{9}{2} + 3e^{-x/200}$ $\Rightarrow \frac{\mathrm{d}}{\mathrm{d}x}\left(ue^{x/200}\right) = \frac{9}{2}e^{x/200} + 3$ $\Rightarrow ue^{x/200} = \ldots$	**M1**	Multiplies their linear equation in standard form by its integrating factor and integrates, at least one RHS term correct. Condone coefficient slips.
	$ue^{x/200} = 900e^{x/200} + 3x + c$	**A1**	Correct, including the constant, oe.
	$x = 0,\ u = 400 \Rightarrow 400 = 900 + c \Rightarrow c = \ldots$	**dM1**	Substitutes $x = 0$, $u = 400$ (or $v = 20$ in v^2) into their solution with a constant to find it. Not for $u = 20$.
	$c = -500 \Rightarrow v^2 = 900 + (3x - 500)e^{-x/200}$	**A1 AG**	Correct c in a correct general solution, reaching the printed result in v^2 with no errors in the part.
		[8]	
10(b)	$\int_0^{200}\left(1800 + 1200e^{-x/200}\right)\mathrm{d}x$ $= \left[1800x - 240\,000e^{-x/200}\right]_0^{200}$	**M1 A1**	**M1** Integrates the driving force with respect to x between 0 and 200, at least one term integrated correctly. **A1** Correct integral; limits need not be substituted yet.
	$= 600\,000 - \frac{240\,000}{e}$	**A1**	Exact oe, or awrt 512 000.
	$x = 200 \Rightarrow v^2 = 900 + \frac{100}{e}$	**B1**	Exact oe, or awrt 937.
	$\frac{1}{2}(800)(20^2) + \text{“}600\,000 - \frac{240\,000}{e}\text{”} - W$ $= \frac{1}{2}(800)\text{“}\left(900 + \frac{100}{e}\right)\text{”}$	**M1 A1**	**M1** Forms an energy equation with both KE terms, their work by the driving force and the work against resistance. Condone sign slips. **A1** Correct equation, oe.
	$W = 400\,000 - \frac{280\,000}{e}$	**A1**	Any exact equivalent.

Question 10 continued

Question	Scheme	Marks	Notes
	$\frac{\text{“}W\text{”}}{200} = 2000 - \frac{1400}{e}$	**B1ft**	ft their work against resistance divided by 200. Exact, or awrt 1480; also accept 1485 (4 s.f.).
		[8]	
	Total for Question 10	**16**	

Question	Scheme	Marks	Notes
11(a)	$2v\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{27}{v} - 3v$	**M1**	Uses Newton's second law with acceleration $v\frac{\mathrm{d}v}{\mathrm{d}x}$, mass 2 and the given force. Condone sign slips.
	$\frac{\mathrm{d}x}{\mathrm{d}v} = \frac{2v^2}{3(9-v^2)}$ $= -\frac{2}{3} + \frac{1}{3-v} + \frac{1}{3+v}$	**A1**	Correct, oe, in any form such as $\frac{2v^2}{3(9-v^2)}$; partial fractions are not needed for this mark.
	$x = \int \left(-\frac{2}{3} + \frac{1}{3-v} + \frac{1}{3+v}\right) \mathrm{d}v$ $= -\frac{2}{3}v - \ln(3-v) + \ln(3+v) + c$	**M1 A1**	**M1** Separates variables and integrates their expression in v, at least one term integrated correctly. Condone sign slips in the logarithms. **A1** Correct, oe; $+c$ may be omitted here.
	$0 = -\frac{2}{3} - \ln 2 + \ln 4 + c \Rightarrow c = \ldots$	**dM1**	Substitutes $x = 0$, $v = 1$ into their integrated equation to find the constant, or uses limits 0 to x and 1 to v.
	$c = \frac{2}{3} - \ln 2$ $\Rightarrow x = \ln\left(\frac{3+v}{2(3-v)}\right) - \frac{2}{3}(v-1)$	**A1 AG**	Correct constant substituted, and log laws leading without error to the printed result; combined steps accepted. The range need not be justified.
		[6]	
11(b)	$x = \ln\frac{5}{2} - \frac{2}{3}$	**B1**	oe, exact.
	$W = \int_1^2 \left(\frac{27}{v} - 3v\right)\frac{2v^2}{3(9-v^2)}\,\mathrm{d}v$ $= \int_1^2 2v\,\mathrm{d}v = \ldots$	**M1**	Writes $\int F\,\mathrm{d}x$ as $\int F\frac{\mathrm{d}x}{\mathrm{d}v}\,\mathrm{d}v$ with their $\frac{\mathrm{d}x}{\mathrm{d}v}$ from (a), or as $\int 2v\,\mathrm{d}v$, and integrates at least one term correctly.
	$W = 3\,\mathrm{J}$	**A1**	
	$F_{\text{av}} = \frac{\text{“3”}}{\text{“}\ln\frac{5}{2} - \frac{2}{3}\text{”}}$	**M1**	Divides their work by their displacement at $v = 2$, exact or decimal.
	$F_{\text{av}} = 12.0\,\mathrm{N}$	**A1**	awrt 12.0.
		[5]	
11(c)	$2\frac{\mathrm{d}v}{\mathrm{d}t} = \frac{27}{v} - 3v$	**M1**	Uses Newton's second law with acceleration $\frac{\mathrm{d}v}{\mathrm{d}t}$, mass 2 and the given force. Condone sign slips.
	$\int \frac{2v}{9-v^2}\,\mathrm{d}v = \int 3\,\mathrm{d}t$ $\Rightarrow -\ln(9-v^2) = 3t + k$	**M1 A1**	**M1** Separates variables and integrates the v side to a logarithm of $9-v^2$, $27-3v^2$ or their factors. Condone a wrong coefficient. **A1** Correct, oe; $+k$ may be omitted here.

Question 11 continued

Question	Scheme	Marks	Notes
	$-\ln(9-1) = 0 + k \Rightarrow k = \ldots$	**dM1**	Substitutes $v = 1$, $t = 0$ into their integrated equation to find the constant, or uses limits 1 to v and 0 to t.
	$k = -\ln 8 \Rightarrow v = \sqrt{9 - 8e^{-3t}}$	**A1**	oe, with v the subject and the positive root only.
		[5]	
11(d)	$v(1) = \sqrt{9 - 8e^{-3}} = 2.933$	**B1ft**	ft their v from (c) at $t = 1$, to 4 s.f.; awrt 2.933 for the correct v.
	$v(2) = \sqrt{9 - 8e^{-6}} = 2.997$	**B1ft**	ft their v from (c) at $t = 2$, to 4 s.f.; awrt 2.997 for the correct v.
	$I = 2\,(\text{“}2.997\text{”} - \text{“}2.933\text{”})$	**M1**	Multiplies 2 by the difference of their two velocities. Condone subtraction in the wrong order.
	$I = 0.128\,\text{N s}$	**A1**	awrt 0.128, positive.
		[4]	
	Total for Question 11	**20**	